CINDY CARMELITA AMANDA

ELECTRONICS ENGINEERING
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PROFESSIONAL SUMMARY

Electronics Engineering graduate (Diploma III) from Politeknik Negeri Semarang with a GPA of 3.69/4.00 (Very Satisfactory), specializing in industrial automation and IoT integration. Strong foundation in electronic circuits, microcontrollers, PLC, control systems, instrumentation, sensors and transducers, and PCB design. Hands-on experience through an industrial automation internship at PT Ruang Industri Indonesia and academic projects ranging from a PLC-based robot with IoT remote control to RFID-based attendance systems. BNSP-certified in Industrial Automation. Also experienced in administrative record keeping and data entry. Skilled in troubleshooting, technical documentation, teamwork, and analytical problem-solving.

EDUCATION

Politeknik Negeri Semarang — **Semarang**
Diploma III in Electronics Engineering | GPA: 3.69 / 4.00 (Very Satisfactory) — ***2023 – 2026***
- Relevant coursework: PLC, Microcontrollers, Internet of Things, Computer-Based Control, Continuous and Discrete Control Systems, Instrumentation, Power Electronics, Electrical Machines.

SMK Negeri 7 Semarang (STEMBA) — **Semarang**
Mechatronics Engineering — ***2018 – 2022***

PROJECT EXPERIENCE

PLC-Based Goods Transport Robot with IoT Remote Control — **Politeknik Negeri Semarang**
Final Project — ***2026***
- Designed a goods-transport robot using a Haiwell AH16S0T PLC and Raspberry Pi 4 Model B with manual (web/remote) and automatic (line-following) modes, driven by a mecanum-drive servo motor system.
- Integrated IoT for real-time remote monitoring and control.
- Applied computer vision (YOLOv8 and ArUco markers) for object detection and precise destination validation.
- Integrated ultrasonic, line-tracking, and load cell sensors; prepared control block diagrams and the technical report.

Encoder-Based DC Motor Position Control Simulation — **Politeknik Negeri Semarang**
Computer-Based Control Laboratory (KBK) Project
- Designed a closed-loop DC motor control system using rotary encoder feedback and a PID algorithm on Arduino Uno with an L293D motor driver.
- Validated the system through virtual hardware simulation in Proteus and a real-time monitoring interface in LabVIEW.

Smart Safe-Deposit Box Based on Arduino Uno — **Politeknik Negeri Semarang**
Microcontroller I Course Project
- Developed a PIN-access safe using Arduino Uno (ATmega328P), keypad, servo motor, I2C LCD, and a buzzer alarm for incorrect PIN entries.

RFID E-Money Toll Gate Prototype Based on Arduino Uno — **Politeknik Negeri Semarang**
Electronics Design and Project Course
- Built a mini toll gate that simulates e-money transactions with an RC522 RFID reader, keypad, LCD, and servo gate, including card balance display and keypad-based top-up.

Automatic Trash Bin with Ultrasonic Sensor — **Politeknik Negeri Semarang**
Group Project
- Developed a touch-free trash bin that opens automatically using an RCWL-1601 ultrasonic sensor and a servo actuator.

IoT-Based Employee Attendance System with RFID — **SMK Negeri 7 Semarang**
Final Project — ***2022***
- Built an attendance system with NodeMCU ESP8266 and an RDM6300 RFID reader that sends data to a database and displays it on a PHP and Bootstrap web interface.
- Implemented an offline mode using a DS3231 RTC and Micro SD card, with automatic data upload once the internet connection is restored.

INTERNSHIP EXPERIENCE

PT Ruang Industri Indonesia
Intern – PLC and Industrial Automation ***September 2025 – December 2025***

- Worked on PLC programming and industrial automation systems.
- Used Visual Studio Code and Python for automation scripting.
- Gained hands-on experience with control systems and sensor integration.

PT DTECH Inovasi Indonesia
Intern ***August 2021 – December 2021***

- Handled inventory management and conducted research and analysis on the Dorlock system and attendance machine.
- Practiced PCB design with Eagle and Arduino IDE.

WORK EXPERIENCE

PT Ungaran Sari Garments
Admin Record ***2022 – 2023***

- Managed administrative records and data entry for production processes.
- Used Microsoft Excel for inventory tracking and reporting.
- Coordinated documentation workflows across departments.

CERTIFICATIONS & TRAINING

- Competency Certificate – Industrial Automation (PLC Programming), Badan Nasional Sertifikasi Profesi (BNSP), 2025. Certificate No.: 78411.31190.0007038 2025
- PersCert TÜV Pneumatic System Maintenance (Pneumatics and Electro-pneumatics), TÜV Rheinland Group. ID: 62372/21/1139
- PCB Design and PCB Manufacturing Training for Industry, INDOMAKER. ID: IIM/ESRT/PCBT/2024/XI/014
- Internship Certificate (PLC, Python, VS Code), PT Ruang Industri Indonesia. ID: MAG/EK/027/001/2026
- Field Work Practice Certificate (Eagle PCB, Arduino IDE), DTECH-Engineering. ID: 13.265/DTECH/XII//2021

LANGUAGE

- English – Intermediate (Professional Reading & Listening, Basic Speaking)
- Indonesian – Native

SKILLS

Technical Skills: PLC Programming, Microcontroller Programming (Arduino), Embedded Systems, Raspberry Pi, Sensors and Transducers, IoT Systems, Control Systems (PID), Electrical Measurement and Instrumentation, Signal Conditioning, Power Electronics, Electrical Machines, PCB Design, Pneumatics and Electro-pneumatics, Maintenance and Repair.

Software Skills: Arduino IDE, HaiwellHappy, CX-Programmer, CX-Designer, Outseal Studio, Node-RED, Proteus, Eagle PCB, EasyEDA, MATLAB, LabVIEW, AutoCAD, FluidSIM, Python, C++, Visual Studio Code, Microsoft Office.

Soft Skills: Technical Communication, Teamwork and Individual Work, Analytical Problem-Solving, Attention to Detail.